Lecture 5: Optimal Portfolios: From Estimation to Decision[1]

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[1] primarily based on Kelly et al. [2023], Brandt et al. [2009]

Review of HF return modeling

- LOB data structure
  - Bid/Ask Price/Volume, deeper levels
- Sampling frequency
  - Trading is not continuous.
- Different clocks
  - calendar, transaction, volume
- Universality
  - OOS, unseen stocks, testing periods, path dependency

From Prediction to Portfolio Construction

Step 1: Return Prediction (Previous Lectures)

We build machine learning models to estimate:

$$\hat{\boldsymbol{\mu}} = \mathbb{E}[\boldsymbol{R}_{t+1} \mid \mathcal{F}_t]$$

Step 2: Decision Problem

Given predicted returns $\hat{\boldsymbol{\mu}}$, how should we allocate wealth?

$$\hat{\boldsymbol{\mu}} \quad \longrightarrow \quad \boldsymbol{w}_t$$

Key Question:

- How do we optimally transform predictions into portfolio weights?
- What objective function should we optimize?

Core Idea: Portfolio construction is an optimization problem that balances:

Expected Return vs. Risk

Heuristic: Long-Short Spread Portfolio

Definition: A strategy that takes long positions in assets with high expected returns and short positions in those with low expected returns.

- **Ranking:** Stocks are sorted into quantiles (e.g., deciles) based on a predictive signal S_i.
- **Spread:** The return is $R_{portfolio} = R_{long} - R_{short}$.
- **Market Neutrality:** Designed to mitigate systematic risk (Market β) and capture pure Alpha.

From Factor Investing to ML:

- *Traditional:* Sorting by a single characteristic (e.g., **Momentum**, **Value**).
- *Machine Learning:* Sorting by a **Composite Score** $\hat{y}_i = f(X_i)$ generated by models.

The Limitation:

- Does it account for the **risk** (variance) of each stock? No.
- Does it account for the **correlation** between assets? No.

Why Optimization Matters

Goal Mismatch:

- **Machine Learning:** Minimizes a statistical loss function (e.g., MSE)

$$\min_{\theta} \mathcal{L}(\theta) = \mathbb{E}[(R_{t+1} - f(X_t; \theta))^2]$$

- **Investors:** Maximize an economic objective (e.g., Mean-Variance Utility)

$$\max_{\boldsymbol{w}} U(\boldsymbol{w}) = \boldsymbol{w}^\top \hat{\boldsymbol{\mu}} - \frac{\gamma}{2} \boldsymbol{w}^\top \hat{\boldsymbol{\Sigma}} \boldsymbol{w}$$

Important Insight:

- A "good" prediction (low MSE) does not always lead to a "good" portfolio (high Sharpe Ratio).
- Optimization is the bridge that aligns statistical accuracy with economic performance.

Mean Variance Portfolio

Mean Variance Portfolio Implementation

Integrated Estimation and Optimization
- Framework 1: Maximum Sharpe Ratio Regression
- Framework 2: Parametric Portfolio Policies

Considering trading costs

Basic properties of portfolio returns

- Portfolio: n-vector of weights indicating the fraction of portfolio wealth held in each asset:

$$\boldsymbol{w} = (w_1, \cdots, w_n)^\top : \sum_{i=1}^{n} w_i = 1$$

- Portfolio return: $R_{\boldsymbol{w}} = \boldsymbol{w}^\top \boldsymbol{R} = \sum_{i=1}^{n} w_i R_i$
- Expected portfolio return $\mu_{\boldsymbol{w}} := \mathbb{E}[R_{\boldsymbol{w}}] = \boldsymbol{w}^\top \boldsymbol{\mu} = \sum_{i=1}^{n} w_i \mu_i$
- Portfolio variance: $\sigma_{\boldsymbol{w}}^2 = \mathrm{Var}[R_{\boldsymbol{w}}] = \mathbb{E}[(R_{\boldsymbol{w}} - \mu_{\boldsymbol{w}})^2] = \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_i \sigma_j \rho_{ij}$ where

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{pmatrix}$$

is the covariance matrix of returns with $\sigma_{ii} = \sigma_i^2$

Warm-up Example

Assume we have two risky assets. Asset 1 has an expected return of 0.1 with a variance of 0.04, and asset 2 has an expected return 0.15 with a variance of 0.09. The covariance between the returns is 0.05.

Question: Calculate the portfolio expected return and variance for the weights $(0.3, 0.7)$.

Warm-up Example

Assume we have two risky assets. Asset 1 has an expected return of 0.1 with a variance of 0.04, and asset 2 has an expected return 0.15 with a variance of 0.09. The covariance between the returns is 0.05.

Question: Calculate the portfolio expected return and variance for the weights $(0.3, 0.7)$.

Answer:

$$\boldsymbol{\mu} = \begin{pmatrix} 0.1 \\ 0.15 \end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix} 0.04 & 0.05 \\ 0.05 & 0.09 \end{pmatrix} \quad \boldsymbol{w} = (0.3, 0.7)^\top$$

$$E[R_{\boldsymbol{w}}] = \boldsymbol{w}^\top \boldsymbol{\mu} = 0.1 \times 0.3 + 0.15 \times 0.7 = 0.135$$

$$\mathrm{Var}[R_{\boldsymbol{w}}] = \boldsymbol{w}^\top \Sigma \boldsymbol{w} = \begin{pmatrix} 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} 0.04 & 0.05 \\ 0.05 & 0.09 \end{pmatrix} \begin{pmatrix} 0.3 \\ 0.7 \end{pmatrix}$$

$$= \begin{pmatrix} 0.047 & 0.078 \end{pmatrix} \begin{pmatrix} 0.3 \\ 0.7 \end{pmatrix} = 0.0687$$

Markowitz Mean-variance Optimization with Risky Assets

Goal: Given n risk assets with $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ such that $\sigma_{ii} > 0$ $(i = 1, 2, \cdots, n)$, evaluate different portfolio $\boldsymbol{w} = (w_1, \cdots, w_n)^\top$ using the mean-variance pair of the portfolio $(\mu_{\boldsymbol{w}}, \sigma^2_{\boldsymbol{w}})$ with preferences for

- higher expected returns $\mu_{\boldsymbol{w}}$
- lower variance $\sigma^2_{\boldsymbol{w}}$

Formulation I: Risk minimization: For a given choice of target mean return $\overline{\mu}$, choose the portfolio $\boldsymbol{w}$ to

$$\begin{aligned} \text{Minimize}: \quad & \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: \quad & \boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu} \\ & \boldsymbol{w}^\top \mathbf{1}_n = 1 \end{aligned}$$

where $\mathbf{1}_n = (1, 1, \cdots, 1)^\top \in \mathbb{R}^n$

Solution: Applying the method of Lagrange multipliers to the convex optimization (minimization) problem subject to linear constraints.

Lagrange Multipliers

The Lagrange multipliers is a strategy for finding the local (or global) maxima and minima of a function subject to equality constraints

- The basic idea is to convert a constrained problem into an unconstrained problem so that the first-order condition (derivative test) can still be applied

Original (constrained) problem:

$$\begin{aligned} \text{Minimize}: &\quad f(\boldsymbol{x}) \text{ with } \boldsymbol{x} \in \mathbb{R}^n \\ \text{Subject to}: &\quad g_j(\boldsymbol{x}) = c_j \ (j = 1, 2, \cdots, m) \end{aligned}$$

Lagrangian function:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \sum_{j=1}^{m} \lambda_j (c_j - g_j(\boldsymbol{x}))$$

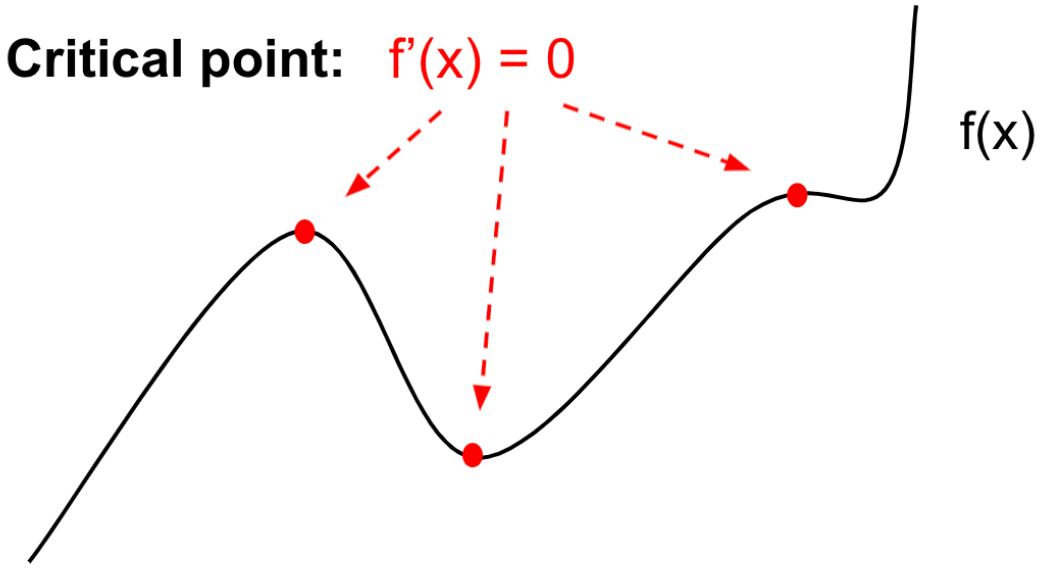
- $\mathcal{L} : \mathbb{R}^{n+m} \to \mathbb{R}$

Lagrange Multipliers

Translated (unconstrained) optimization problem:

$$\begin{cases} \frac{\partial}{\partial x_i}\mathcal{L}(\boldsymbol{x},\boldsymbol{\lambda}) = \frac{\partial}{\partial x_i}f(\boldsymbol{x}) - \sum_{j=1}^{m}\lambda_j\frac{\partial}{\partial x_i}g_j(\boldsymbol{x}) = \mathbf{0}, \ (i = 1, 2, \cdots, n) \\ \frac{\partial}{\partial \lambda_j}\mathcal{L}(\boldsymbol{x},\boldsymbol{\lambda}) = c_j - g_j(\boldsymbol{x}) = 0, \ (j = 1, 2, \cdots, m) \end{cases}$$

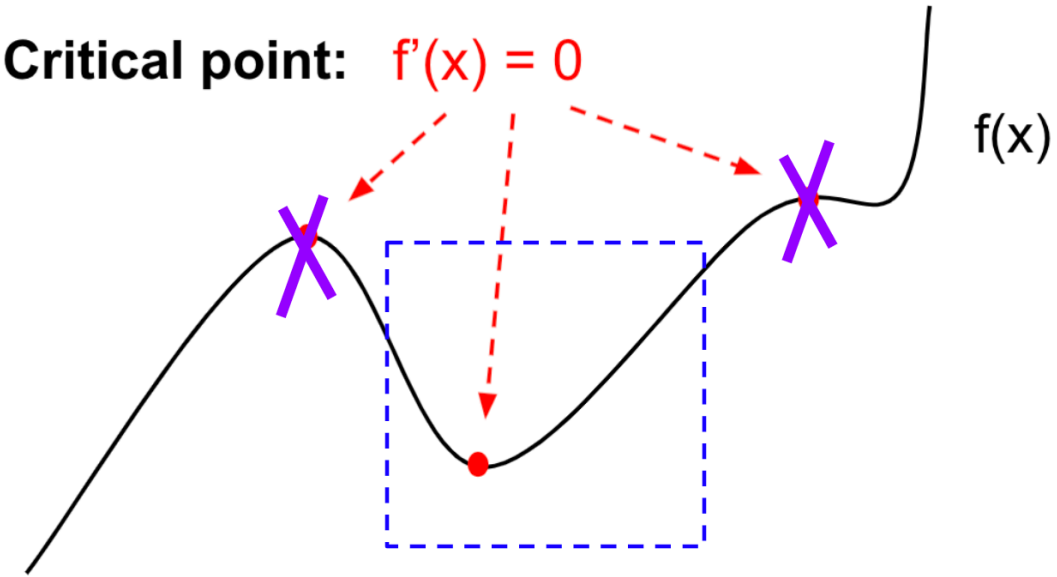
- (m+n) unknowns and (m+n) equation*s
- Any solution of this system $(\boldsymbol{x}_*; \boldsymbol{\lambda}_*)$ is called a critical point of the Lagrange function $\mathcal{L}$



Method Of Lagrange Multipliers

Sufficient condition for global extrema: Let $(\boldsymbol{x}_*, \boldsymbol{\lambda}_*)$ be a critical point of $\mathcal{L}$ of optimization problem, If $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}_*)$ is convex in $\boldsymbol{x}$, then $\boldsymbol{x}_*$ is a global minimum of $f(\boldsymbol{x})$ subject to $g_j(\boldsymbol{x}) = c_j$ $(j = 1, 2, \cdots, m)$.

- ⋆ This sufficient condition is enough for the mean-variance optimization problem!
- **For Quadratic Programming:** Since $f(\boldsymbol{x})$ is convex (if $\boldsymbol{\Sigma}$ is PSD) and constraints are linear, the critical point is the **unique global minimum**.



Risk Minimization Without Budget Constraint

$$\begin{aligned} \text{Minimize}: \quad & \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: \quad & \boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu} \end{aligned}$$

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- Define the Lagrangian $\mathcal{L}(\boldsymbol{w}, \lambda_1) = \frac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} + \lambda_1(\overline{\mu} - \boldsymbol{w}^\top \boldsymbol{\mu})$
- Derive the first-order conditions

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \boldsymbol{w}} &= \boldsymbol{0}_n = \boldsymbol{\Sigma} \boldsymbol{w} - \lambda_1 \boldsymbol{\mu} \\ \frac{\partial \mathcal{L}}{\partial \lambda_1} &= 0 = \overline{\mu} - \boldsymbol{w}^\top \boldsymbol{\mu} \end{aligned}$$

- Solve for $\boldsymbol{w}$ in terms of λ_1:

$$\boldsymbol{w}_* = \lambda_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$$

Final Solution for Portfolio Weights

Using the return constraint $\boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu}$:

$$(\lambda_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu})^\top \boldsymbol{\mu} = \overline{\mu}$$

This simplifies to:

$$\lambda_1 \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} = \overline{\mu}$$

Thus, the Lagrange multiplier λ_1 is:

$$\lambda_1 = \frac{\overline{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}}$$

Substituting λ_1 back into the equation* for $\boldsymbol{w}$,

$$\boldsymbol{w}_* = \frac{\overline{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$$

- The portfolio weights are proportional to $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, which reflects the trade-off between risk (via $\boldsymbol{\Sigma}^{-1}$) and expected return (via $\boldsymbol{\mu}$).
- The proportionality factor $\frac{\overline{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}}$ ensures that the portfolio achieves the target return $\overline{\mu}$.

Global Minimum Variance Portfolio (GMVP)

The GMVP focuses solely on minimizing portfolio risk, completely ignoring expected returns. It is formulated as the following convex quadratic program:

$$\begin{array}{ll} \underset{\mathbf{w}}{\text{minimize}} & \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \\ \text{subject to} & \mathbf{1}^T \mathbf{w} = 1 \end{array}$$

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The optimal solution to this problem is given by:

$$\mathbf{w}_{\text{GMVP}} = \frac{1}{\mathbf{1}^T \boldsymbol{\Sigma}^{-1} \mathbf{1}} \boldsymbol{\Sigma}^{-1} \mathbf{1}$$

Due to its simplicity, the GMVP is widely used in academic research to evaluate and compare different covariance matrix estimators, while *avoiding the complexities of estimating expected returns*.

Risk Minimization Problem

$$\begin{aligned} \text{Minimize}: &\quad \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: &\quad \boldsymbol{w}^\top \boldsymbol{\mu} = \overline{\mu}, \boldsymbol{w}^\top \mathbf{1}_n = 1 \end{aligned}$$

- Define the Lagrangian
  $\mathcal{L}(\boldsymbol{w}, \lambda_1, \lambda_2) = \frac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} + \lambda_1(\overline{\mu} - \boldsymbol{w}^\top \boldsymbol{\mu}) + \lambda_2(1 - \boldsymbol{w}^\top \mathbf{1})$
- Derive the first-order conditions

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \boldsymbol{w}} &= \mathbf{0}_n = \boldsymbol{\Sigma} \boldsymbol{w} - \lambda_1 \boldsymbol{\mu} - \lambda_2 \mathbf{1}_n \\ \frac{\partial \mathcal{L}}{\partial \lambda_1} &= 0 = \overline{\mu} - \boldsymbol{w}^\top \boldsymbol{\mu} \\ \frac{\partial \mathcal{L}}{\partial \lambda_2} &= 0 = 1 - \boldsymbol{w}^\top \mathbf{1}_n \end{aligned}$$

- Solve for $\boldsymbol{w}$ in terms of λ_1, λ_2:

$$\boldsymbol{w}_* = \lambda_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} + \lambda_2 \boldsymbol{\Sigma}^{-1} \mathbf{1}_n$$

Full MVO: Dual Constraints Solution

Geometric Interpretation:

- The optimal portfolio $\boldsymbol{w}_*$ lies in the span of two specific vectors: $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}^{-1}\mathbf{1}_n$.
- $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$: Focuses on tilting towards assets with high expected returns (adjusted by covariance).
- $\boldsymbol{\Sigma}^{-1}\mathbf{1}_n$: Focuses on minimizing total variance (GMVP).

Deep Dive: How do we find λ_1, λ_2?

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Deep Dive: How do we find λ_1, λ_2?
We substitute $\boldsymbol{w}_*$ back into the two constraints, leading to a 2×2 linear system involving the scalar products $\boldsymbol{\mu}^\top \Sigma^{-1}\boldsymbol{\mu}$, $\boldsymbol{\mu}^\top \Sigma^{-1}\mathbf{1}$, and $\mathbf{1}^\top \Sigma^{-1}\mathbf{1}$.

Risk Minimization Problem

- Solve for λ_1 and λ_2 by substituting for $\boldsymbol{w}^*$:

$$\begin{aligned}
\overline{\mu} &= \boldsymbol{w}_*^\top \boldsymbol{\mu} = \lambda_1(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}) + \lambda_2(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) \\
1 &= \boldsymbol{w}_*^\top \mathbf{1}_n = \lambda_1(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) + \lambda_2(\mathbf{1}_n^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) \\
\Rightarrow \quad & \begin{pmatrix} \overline{\mu} \\ 1 \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} \text{ with}
\end{aligned}$$

$a = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $b = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n$, and $c = \mathbf{1}_n^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n$.
Hence

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} \begin{pmatrix} \overline{\mu} \\ 1 \end{pmatrix}.$$

Risk Minimization Problem

Variance of optimal portfolio with return $\overline{\mu}$: With the given values of λ_1 and λ_2, the solution portfolio $\boldsymbol{w}_* = \lambda_1 \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \lambda_2 \boldsymbol{\Sigma}^{-1}\mathbf{1}_n$ has minimum variance equal to

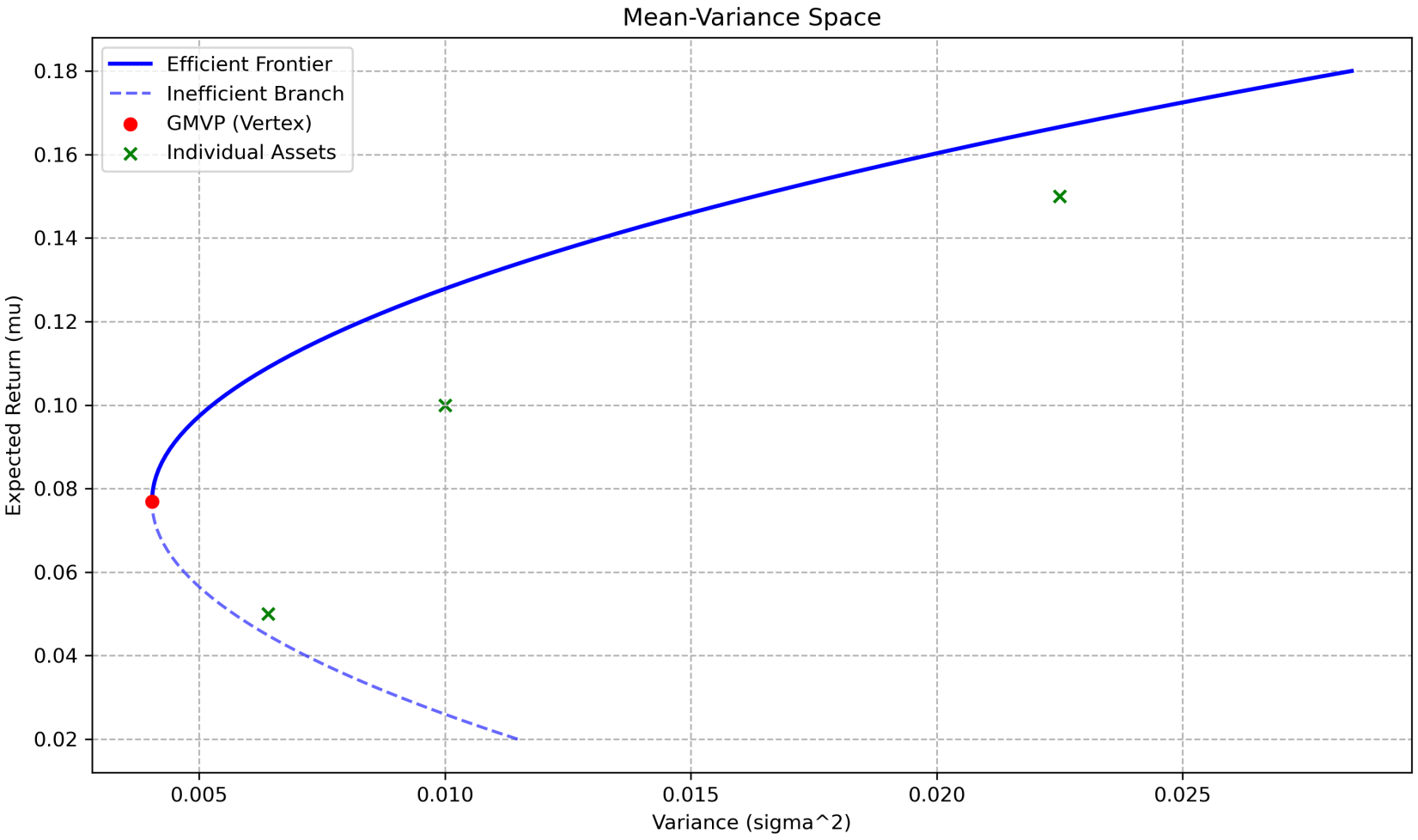
$$\begin{aligned}
\sigma_*^2 &= \boldsymbol{w}_*^\top \boldsymbol{\Sigma} \boldsymbol{w}_* \\
&= \lambda_1^2(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}) + 2\lambda_1\lambda_2(\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) + \lambda_2^2(\mathbf{1}_n^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}_n) \\
&= (\lambda_1, \lambda_2) \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix}
\end{aligned}$$

Substituting $\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} \begin{pmatrix} \overline{\mu} \\ 1 \end{pmatrix}$ gives

$$\boxed{\sigma_*^2 = (\overline{\mu}, 1) \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} \begin{pmatrix} \overline{\mu} \\ 1 \end{pmatrix} = \frac{1}{ac - b^2}(c\overline{\mu}^2 - 2b\overline{\mu} + a)}$$

Efficient Frontier

- Optimal portfolio has variance σ_*^2: parabolic in the mean $\bar{\mu}$



Other Formulations

Formulation II: Expected Return Maximization. For a given choice of target return variance $\overline{\sigma}^2$, choose the portfolio $\boldsymbol{w}$ to

$$\begin{aligned} \text{Maximize}: \quad & \boldsymbol{w}^\top \boldsymbol{\mu} \\ \text{Subject to}: \quad & \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} = \overline{\sigma}^2 \\ & \boldsymbol{w}^\top \mathbf{1}_n = 1 \end{aligned}$$

Formulation III: Risk Aversion Optimization. Let $\lambda \geq 0$ denote the risk aversion preference gauging the trade-off between risk and return. Choose the portfolio $\boldsymbol{w}$ to

$$\begin{aligned} \text{Maximize}: \quad & \mathbb{E}[R_{\boldsymbol{w}}] - \tfrac{\lambda}{2}\mathrm{Var}[R_{\boldsymbol{w}}] = \boldsymbol{w}^\top \boldsymbol{\mu} - \tfrac{\lambda}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: \quad & \boldsymbol{w}^\top \mathbf{1}_n = 1 \end{aligned}$$

Mean-variance with (Additional) Risk-free Asset

Risk-free asset: In addition to the risky asset ($i = 1, 2, \cdots, n$), we assume there is a risk-free asset ($i = 0$) for which

$$R_0 = \mu_0, \text{ i.e., } \mathbb{E}[R_0] = \mu_0, \ \mathrm{Var}[R_0] = 0$$

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Portfolio with investment in risk-free asset

- Suppose the investor can invest in the n risky investment as well as in the risk-free asset: $\boldsymbol{w}^\top \mathbf{1}_n = \sum_{i=1}^n w_i$ is invested in risky assets and $1 - \boldsymbol{w}^\top \mathbf{1}_n$ is invested in the risk-free asset.
- If borrowing is allowed, $(1 - \boldsymbol{w}^\top \mathbf{1}_n)$ can be negative
- Portfolio: $R_{\boldsymbol{w}} = \boldsymbol{w}^\top \boldsymbol{R} + (1 - \boldsymbol{w}^\top \mathbf{1}_n) R_0$, where $\boldsymbol{R} = (R_1, \cdots, R_n)$, has expected return and variance

$$\begin{aligned} \mu_{\boldsymbol{w}} &= \boldsymbol{w}^\top \boldsymbol{\mu} + (1 - \boldsymbol{w}^\top \mathbf{1}_n)\mu_0 \\ \sigma_{\boldsymbol{w}}^2 &= \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \end{aligned}$$

Note: R_0 has zero variance and is uncorrelated with $\boldsymbol{R}$.

Mean-variance Optimization with Risk-free Asset

Formulation I′: Risk optimization with risk-free asset For a given choice of target mean return $\overline{\mu}$, choose the portfolio $\boldsymbol{w}$ to

$$\begin{aligned} \text{Minimize}: \quad & \tfrac{1}{2}\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{Subject to}: \quad & \boldsymbol{w}^\top \boldsymbol{\mu} + (1 - \boldsymbol{w}^\top \mathbf{1}_n)\mu_0 = \overline{\mu} \end{aligned}$$

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Solution: Apply the method of Lagrange multipliers to the convex optimization (minimization):

- Define the Lagrangian

$$\mathcal{L}(\boldsymbol{w}, \lambda_1) = \frac{1}{2}\boldsymbol{w}\boldsymbol{\Sigma}\boldsymbol{w} + \lambda_1[(\overline{\mu} - \mu_0) - \boldsymbol{w}^\top(\boldsymbol{\mu} - \mu_0 \mathbf{1}_n)]$$

- Derive the first-order conditions

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Mean-variance Optimization with Risk-free Asset

Solution (cont):

- Solve for $\mathbf{w}$ in terms of λ_1:

$$\mathbf{w}_* = \lambda_1 \Sigma^{-1} \left[\mu - \mathbf{1}_n \mu_0 \right]$$

$$\text{and} \quad \lambda_1 = \frac{\overline{\mu} - \mu_0}{(\mu - \mathbf{1}_n \mu_0)^\top \Sigma^{-1} (\mu - \mathbf{1}_n \mu_0)}$$

Mean-variance Optimization with Risk-free Asset

Exercise. Asset 1 has an expected return of 10% with a standard deviation of 25%, and asset 2 has an expected return of 15% and a standard deviation of 35%. The covariance between the returns is 0.0175 and the risk-free rate is 8%.

(a) What is the optimal portfolio consisting of risky assets and risk-free asset if you want an average return of 0.10?

Mean-variance Optimization with Risk-free Asset

Exercise. Asset 1 has an expected return of 10% with a standard deviation of 25%, and asset 2 has an expected return of 15% and a standard deviation of 35%. The covariance between the returns is 0.0175 and the risk-free rate is 8%.

(a) What is the optimal portfolio consisting of risky assets and risk-free asset, if you want an average return of 0.10?

Solution: Note that

$$\Sigma = \begin{pmatrix} 0.0625 & 0.0175 \\ 0.0175 & 0.1225 \end{pmatrix}, \quad \mu = \begin{pmatrix} 0.10 \\ 0.15 \end{pmatrix}, \quad \overline{\mu} = 0.1, \quad \mu_0 = 0.08,$$

and we have the formula

$$\mathbf{w}_* = \frac{(\overline{\mu} - \mu_0)\,\Sigma^{-1}(\mu - \mathbf{1}_n\mu_0)}{(\mu - \mathbf{1}_n\mu_0)^\top \Sigma^{-1}(\mu - \mathbf{1}_n\mu_0)}. \tag{1}$$

First, we have

$$\Sigma^{-1} = \frac{1}{0.0625 \times 0.1225 - 0.0175 \times 0.0175} \begin{pmatrix} 0.1225 & -0.0175 \\ -0.0175 & 0.0625 \end{pmatrix}.$$

Mean-variance Optimization with Risk-free Asset

Solution (cont I): and

$$\mu - \mathbf{1}_n \mu_0 = \begin{pmatrix} 0.10 \\ 0.15 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} \times 0.08 = \begin{pmatrix} 0.02 \\ 0.07 \end{pmatrix}.$$

Then

$$(\overline{\mu} - \mu_0)\, \Sigma^{-1} (\mu - \mathbf{1}_n \mu_0) \;=\; \frac{(0.10 - 0.08)}{0.0625 \times 0.1225 - 0.0175 \times 0.0175} \begin{pmatrix} 0.1225 & -0.0175 \\ -0.0175 & 0.0625 \end{pmatrix} \begin{pmatrix} 0.02 \\ 0.07 \end{pmatrix}$$

$$= 2.7210884 \times \begin{pmatrix} 0.001225 \\ 0.004025 \end{pmatrix}.$$

and

$$(\mu - \mathbf{1}_n \mu_0)^\top \Sigma^{-1} (\mu - \mathbf{1}_n \mu_0) = \frac{1}{0.0625 \times 0.1225 - 0.0175 \times 0.0175}$$

$$\times\, (0.02,\, 0.07) \begin{pmatrix} 0.1225 & -0.0175 \\ -0.0175 & 0.0625 \end{pmatrix} \begin{pmatrix} 0.02 \\ 0.07 \end{pmatrix} = 0.04166.$$

Mean-variance Optimization with Risk-free Asset

Solution (cont II):
Hence we have

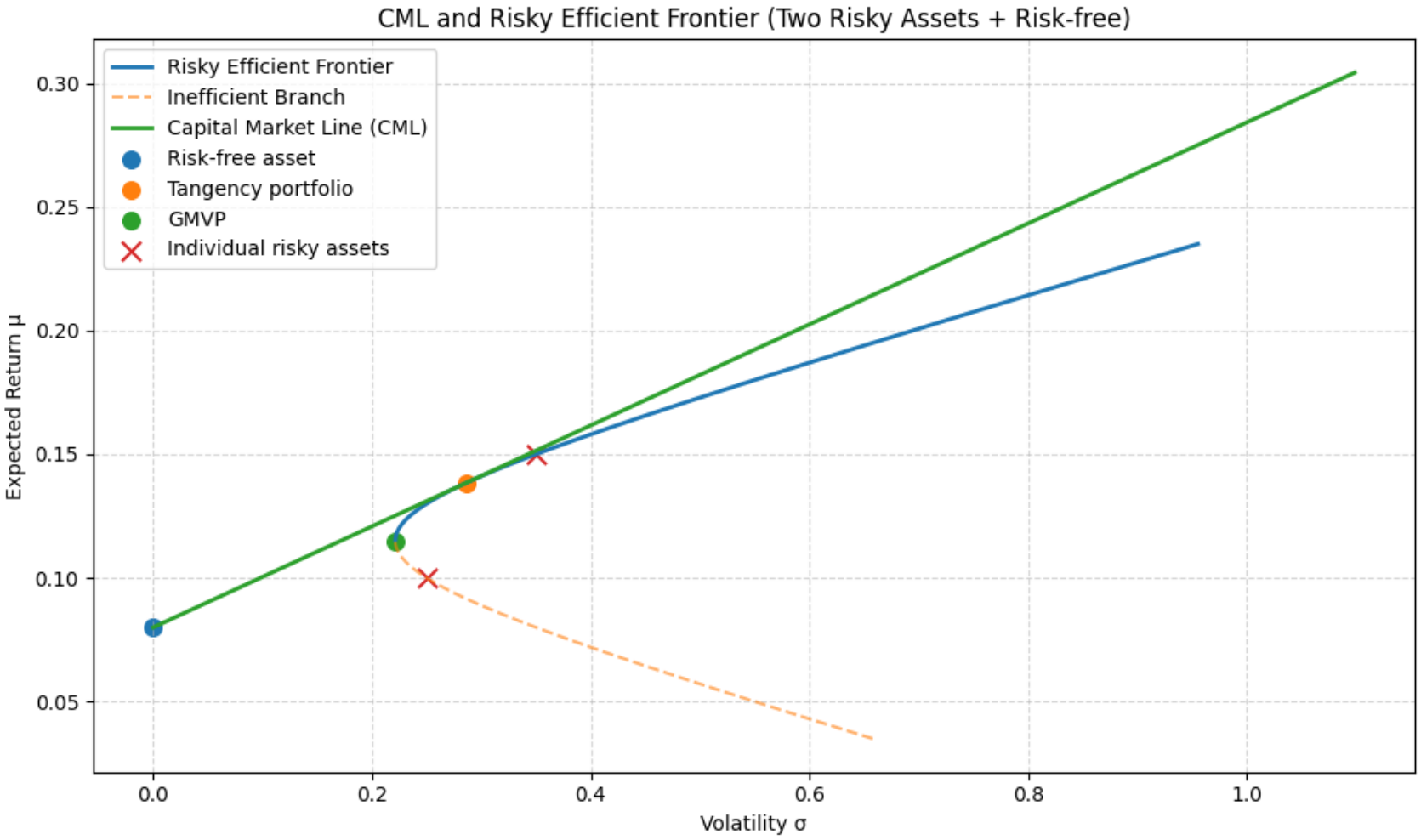
$$\mathbf{w}_* = \frac{(\overline{\mu} - \mu_0)\,\Sigma^{-1}(\mu - \mathbf{1}_n\mu_0)}{(\mu - \mathbf{1}_n\mu_0)^\top \Sigma^{-1}(\mu - \mathbf{1}_n\mu_0)} = \frac{2.7210884}{0.04166}\begin{pmatrix}0.001225\\0.004025\end{pmatrix} = \begin{pmatrix}0.08\\0.26\end{pmatrix},$$

and

$$w_0 = 1 - 0.08 - 0.26 = 0.66$$

is the proportion invested in the risk-free asset.

(b) What is the optimal portfolio consisting of only risky assets, if you want an average return of 0.10?



Practical constraints 1

Capital Budget Constraint: Ensures full capital allocation.

$$\mathbf{1}^T \mathbf{w} = 1$$

Long-Only Constraint: No short selling allowed.

$$\mathbf{w} \geq 0$$

Dollar-Neutral (Self-Financing) Constraint: No additional capital investment.

$$\mathbf{1}^T \mathbf{w} = 0$$

Holding Constraint: Restricts asset positions within limits.

$$\mathbf{l} \leq \mathbf{w} \leq \mathbf{u}$$

Practical constraints 2

Leverage Constraint: Limits total exposure to leverage limit L.

$$\|\mathbf{w}\|_1 \leq L$$

Cardinality Constraint: Restricts the number of selected assets.

$$\|\mathbf{w}\|_0 \leq K$$

- an investor wishes to hold only a small subset of available assets (e.g., due to transaction costs or simplicity).

Turnover Constraint: Controls changes in asset holdings.

$$\|\mathbf{w} - \mathbf{w}_0\|_1 \leq u$$

where $\mathbf{w}_0$ is the currently held portfolio.

- institutional investors aim to maintain stable holdings over time.

Market-Neutral Constraint: Ensures beta neutrality.

$$\boldsymbol{\beta}^T \mathbf{w} = 0$$

- $\boldsymbol{\beta}$ represents the vector of factor loadings (e.g., stock betas with respect to the market).

Dollar-Neutral vs. Market-Neutral

Dollar-Neutral (DN)

- **Focus:** Capital allocation.
- **Constraint:** $\mathbf{1}^T\mathbf{w} = 0$.
- **Goal:** Self-financing; long positions equal short positions in dollar value.
- **Risk:** Does not guarantee zero risk if asset betas differ.

Market-Neutral (MN)

- **Focus:** Risk exposure.
- **Constraint:** $\boldsymbol{\beta}^T\mathbf{w} = 0$.
- **Goal:** Zero correlation with the market; capture pure Alpha.
- **Risk:** Hedged against systematic (market) movements.

Case Study: Dollar-Neutral $\neq$ Market-Neutral

Hypothetical Scenario:

- **Long:** \$1.0M in Stock A ($\beta_A = 1.5$)
- **Short:** \$1.0M in Stock B ($\beta_B = 0.5$)

The Result: Dollar-Neutral

Net Dollar Exposure $= 0$.
The portfolio appears "balanced" on paper.

The Reality: NOT Market-Neutral

Net Portfolio
$\beta = (1.5 - 0.5) = 1.0$.
If market drops 10%, portfolio loses $\sim$10%.

Optimization Insight: To reach **Market-Neutrality**, we must solve for $\sum w_i \beta_i = 0$. In this case, we would need to short \$3.0M of Stock B for every \$1.0M of Stock A.

Mean Variance Portfolio

Mean Variance Portfolio Implementation

Integrated Estimation and Optimization
- Framework 1: Maximum Sharpe Ratio Regression
- Framework 2: Parametric Portfolio Policies

Considering trading costs

Estimation

- Markowitz (1952) framework assumes full knowledge of return distributions.
- Real-world investors make portfolio decisions under uncertainty.
- $\mathcal{F}_{t-1}$ denotes the available historical data up to time $t-1$.
- $\mathbf{r}_t$ follows a multivariate stochastic process with:
  - **Conditional mean:**

$$\boldsymbol{\mu}_t \triangleq \mathrm{E}\left[\mathbf{r}_t \mid \mathcal{F}_{t-1}\right]$$

  - **Conditional covariance:**

$$\boldsymbol{\Sigma}_t \triangleq \mathrm{Cov}\left[\mathbf{r}_t \mid \mathcal{F}_{t-1}\right] = \mathrm{E}\left[\left(\mathbf{r}_t - \boldsymbol{\mu}_t\right)\left(\mathbf{r}_t - \boldsymbol{\mu}_t\right)^T \mid \mathcal{F}_{t-1}\right]$$

- These moments $(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ play a crucial role portfolio optimization.

Parameter Estimation

- Independent and Identically Distributed (i.i.d.) Model:

$$\mathbf{r}_t = \boldsymbol{\mu} + \epsilon_t$$

  - $\boldsymbol{\mu} \in \mathbb{R}^N$ is the mean return vector.
  - $\epsilon_t \in \mathbb{R}^N$ is an i.i.d. noise process with zero mean and constant covariance $\boldsymbol{\Sigma}$.
- **Estimation:** $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ need to be estimated from T observations.
- **Basic Sample Estimators:**
  - Sample mean: $\hat{\boldsymbol{\mu}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{r}_t$
  - Sample covariance matrix: $\hat{\boldsymbol{\Sigma}} = \frac{1}{T-1}\sum_{t=1}^{T}(\mathbf{r}_t - \hat{\boldsymbol{\mu}})(\mathbf{r}_t - \hat{\boldsymbol{\mu}})^T$

Plug-in Portfolios

- Investors often estimate return moments first, then optimize portfolio weights.
- The "plug-in" estimator replaces unknown parameters with estimated values.

$$\widehat{w} = \frac{1}{\gamma}\widehat{\Sigma}^{-1}\widehat{\mu}$$

Limitations:

- Consistency of estimates does not guarantee good finite-sample performance.
- Plug-in portfolios perform poorly when N is large relative to T.
- High dimensional settings exacerbate estimation errors, leading to suboptimal decisions.

Economic Loss from Estimation Error

In Kan and Zhou [2007], the difference in investor utility between the true optimal portfolio w^* and the plug-in portfolio $\widehat{w}$ is treated as an economic loss function:

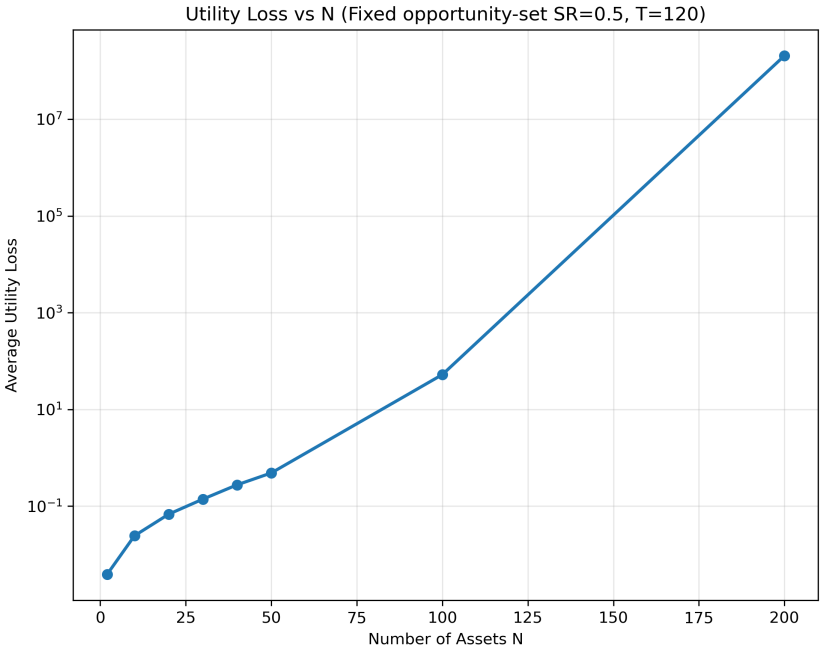
$$\mathrm{E}\left[L\left(w^*, \widehat{w}\right) \mid \mu, \Sigma\right] = U\left(w^*\right) - \mathrm{E}[U(\widehat{w}) \mid \mu, \Sigma]$$

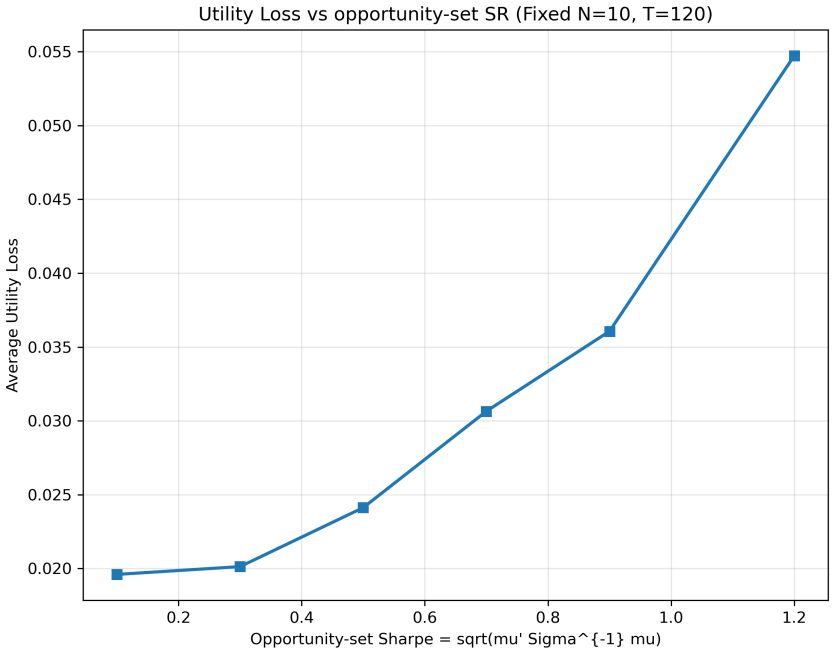
- Statistical "risk function"
- Utility loss from using the plug-in portfolio can be quantified:

$$\mathrm{E}\left[L\left(w^*, \widehat{w}\right) \mid \mu, \Sigma\right] = a_1 \frac{\mathrm{SR}^2}{2\gamma} + a_2$$

- Loss increases with the number of assets N and Sharpe ratio.
- Investors experience larger utility costs in high-dimensional settings.

Utility Loss





- Don't be fooled by dimensionality: More assets $\neq$ More realized utility (unless $T \gg N$).
- Don't be fooled by high returns: High potential SR $\neq$ Easy profit (it requires higher precision in estimation).

Challenges in Parameter Estimation

1. **Estimation Error Scales with Dimension**
   - When N/T is non-negligible, plug-in portfolios become unstable.
   - The sample mean has extremely low signal-to-noise ratio in financial data.
2. **Increasing T is Not a Free Lunch**
   - Longer samples reduce variance but increase model misspecification risk.
   - Financial returns violate the i.i.d. assumption (structural breaks, regime shifts).
3. **Optimization Amplifies Noise**
   - Inverting $\hat{\Sigma}$ magnifies small eigenvalue errors.
   - Result: extreme weights and poor out-of-sample performance.
4. **Practical Implication**
   - Naive plug-in mean–variance optimization is rarely used without regularization or shrinkage.

Structural Limitations of MVO

1. **Variance as a Risk Measure**
   - Variance penalizes both downside and upside deviations.
   - Investors are typically concerned only with downside risk.
   - Alternatives:
     - Value at Risk (VaR)
     - Conditional Value at Risk (CVaR)
2. **Extreme and Concentrated Portfolios**
   - Optimal weights often load heavily on a few assets.
   - Small changes in inputs lead to large reallocations.
   - The solution lacks stability and diversification.
3. **Quadratic Utility Assumption**
   - Implicitly assumes symmetric preferences.
   - Not consistent with fat tails and skewness in returns.

Mitigating Plug-in Portfolio Issues

1. **Improve Parameter Estimation**
   - Mean shrinkage (Bayes–Stein)
   - Covariance shrinkage (Ledoit–Wolf)
   - Factor models and high-dimensional covariance estimators
   - Black–Litterman
2. **Stabilize the Optimization Problem**
   - Portfolio constraints (weight limits, turnover constraints)
   - Regularization (L2 / L1 penalties)
   - Robust optimization
3. **Integrated Estimation and Decision**
   - Integrated estimation and utility optimization improves performance.

Mean Variance Portfolio

Mean Variance Portfolio Implementation

Integrated Estimation and Optimization
- Framework 1: Maximum Sharpe Ratio Regression
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Considering trading costs

Quadratic utility + i.i.d. Normality

- Under simple assumptions (i.i.d. normal returns, quadratic utility), the optimal portfolio should depend on sample statistics:
$$\widehat{w} = f(\widehat{\mu}, \widehat{\Sigma})$$
- Kan and Zhou [2007] show that certain linear modifications improve the plug-in estimator.
  - mixing with plug-in minimum variance portfolio
- good choices for $\widehat{w}$ "can potentially be a very complex nonlinear function of $\widehat{\mu}$ and $\widehat{\Sigma}$ and there can be infinitely many ways to construct it. However, it is not an easy matter to determine the optimal $f(\widehat{\mu}, \widehat{\Sigma})$."

Integrated Estimation and Optimization

- Unknown function f can be parameterized flexibly, e.g., using neural networks.
- The goal is to optimize expected out-of-sample utility while managing estimation uncertainty.

Machine Learning Workflow for Portfolio Choice:

Step 1: Model Training

- Choose a function class indexed by a tuning parameter (e.g., ridge penalty z).
- Estimate model parameters using the training sample.

Step 2: Validation and Regularization

- Select tuning parameters that optimize expected out-of-sample performance.
- Use cross-validation to approximate expected utility without closed-form solutions.

Generalizing Portfolio Choice with Machine Learning

- The portfolio rule can be written as:

$$\widehat{w} = f_\theta\left(X_T\right)$$

- X_T can include past returns, macroeconomic indicators, and other conditioning variables.
- the return distribution need not be normal nor i.i.d.
- Flexible machine learning methods allow customization for tail risk, multi-horizon objectives, and alternative return distributions.
- Integrated Always Better?

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- the return distribution need not be normal nor i.i.d.
- Flexible machine learning methods allow customization for tail risk, multi-horizon objectives, and alternative return distributions.
- Integrated Always Better?
- Two-step assumes we know the model and just need to estimate it. Integrated assumes the decision rule itself should be learned.

Maximum Sharpe Ratio as OLS Regression

Markowitz plug-in solution (excess returns):

$$w^{\text{OLS}} \propto \widehat{\Sigma}^{-1}\widehat{\mu}$$

Equivalent Regression Problem:

Consider the OLS regression

$$1 = w' R_t + u_t$$

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Equivalent Regression Problem:
Consider the OLS regression

$$1 = w' R_t + u_t$$

This corresponds to solving:

$$\min_{w} \frac{1}{T}\sum_{t=1}^{T}(1 - w' R_t)^2$$

First-Order Condition:

$$\widehat{\Sigma} w = \widehat{\mu}$$

Thus,

$$\widehat{w} = \widehat{\Sigma}^{-1}\widehat{\mu}$$

Key Insight: Maximum Sharpe ratio portfolio can be obtained via OLS.

Economic Interpretation of MSRR

Interpretation:
We seek a combination of risky returns R_t that behaves as closely as possible to a positive constant.

Why does this maximize Sharpe ratio?

- High mean $\rightarrow$ portfolio return close to 1 on average
- Low variance $\rightarrow$ small regression residuals

Integrated Perspective:
Instead of estimating $\hat{\mu}$ and $\hat{\Sigma}$ separately,

$$\hat{w} = \arg\min_{w} \frac{1}{T} \sum (1 - w' R_t)^2$$

This directly optimizes the portfolio objective.

Key Insight:
Portfolio choice can be reformulated as a regression problem — opening the door to machine learning methods.

Example 1: LASSO-Regularized MSRR

Add L1 regularization:

$$\min_{w} \frac{1}{T} \sum (1 - w' R_t)^2 + \lambda \|w\|_1$$

Economic Effects:

- Controls leverage
- Induces sparsity
- Reduces estimation error amplification

LASSO or RIDGE?

Example 1: LASSO-Regularized MSRR

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Economic Effects:

- Controls leverage
- Induces sparsity
- Reduces estimation error amplification

LASSO or RIDGE?

- RIDGE is Covariance Shrinkage

Example 1: LASSO in MSRR

Ao et al. [2019] proposed:

$$\widehat{\boldsymbol{w}^*} = \arg\min_{\boldsymbol{w}} ||\widehat{\boldsymbol{r}_c} - \boldsymbol{R}\boldsymbol{w}||_2^2 \quad \text{subject to } ||\boldsymbol{w}||_1 \leq \lambda,$$

Table 6

Summary of risk, Sharpe ratio and p-value of Sharpe ratio test of the portfolios under comparison on DJIA 30 index constituents and Fama-French three factors. The testing period is February 1967 – December 2016. The risk constraint $\sigma = 0.0366$ is the standard deviation of the excess returns on DJIA 30 index during February 1962 – January 1967.

DJIA 30 & FF3	σ=0.0366	$T = 60$	
Portfolio	Risk	Sharpe Ratio	p-value
Index	0.044	0.102	0.000
Equally weighted	0.043	0.064	0.000
Factor	0.040	0.425	0.000
KZ	0.124	0.535	0.000
MAXSER	0.044	**0.701**	–

Example 2: Dynamic Portfolio Weight Parameterization

Parametric Portfolio Rule:

$$w_t = S_{t-1}\beta$$

- S_{t-1}: $N \times K$ matrix of signals
- β: $K \times 1$ vector of signal weights

Interpretation:

- Asset weights vary with conditioning information
- Portfolio is linear in signals

Substituting into MSRR:

$$1 = \beta'(S'_{t-1}R_t) + u_t$$

Define:

$$F_t = S'_{t-1}R_t$$

We now run MSRR on K factor returns F_t.

From Assets to Signal-Managed Portfolios

Step 1: Construct factor returns

$$F_t = S'_{t-1} R_t$$

- Each column of S defines a characteristic-managed portfolio

Step 2: Apply MSRR at factor level

$$\hat{\beta} = \widehat{\mathrm{Cov}}(F)^{-1} \widehat{E}(F)$$

Advantages:

- Dimension reduction ($K \ll N$)
- Lower estimation error
- Direct testing of signal contributions (OLS statistics)

Advantages of the MSRR-Parametric Framework

1. **Decision-Focused Formulation**

$$\min_{\beta} \frac{1}{T} \sum_{t=1}^{T} (1 - \beta' F_t)^2$$

   - Directly maximizes Sharpe ratio
   - No need to separately estimate μ and Σ

2. **Dimension Reduction**
   - Operates on K signal-managed portfolios instead of N assets
   - Reduces estimation error when $K \ll N$

3. **Machine Learning Compatibility**
   - Large K with rich signals and interactions
   - Regularization (Ridge, LASSO, Elastic Net)
   - Easy incorporation of leverage constraints (L_1 penalty)

High Complexity MSRR with Neural Networks

- MSRR can be extended using machine learning techniques such as neural networks.
- Nonlinear transformations improve flexibility in portfolio construction.
- Signals / Representation:

$$S_{i,t}(\theta) = q\left(X_{i,t}; \theta\right)$$

- Portfolio weights / Decision:

$$w_{i,t} = \beta' S_{i,t}(\theta)$$

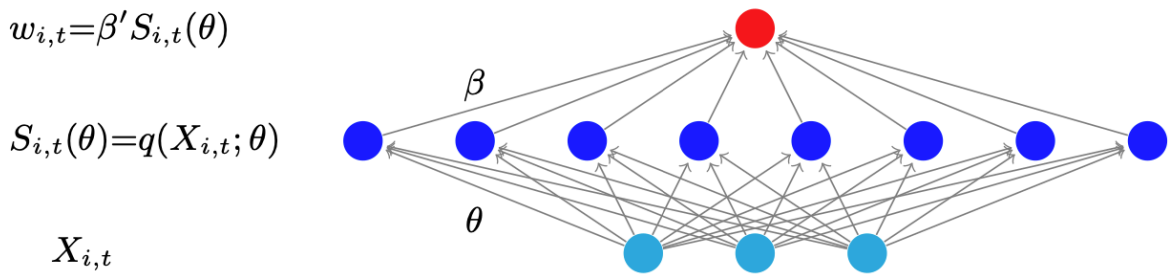

Figure 5.1: MSRR Neural Network

Note: Illustration of a simple neural architecture mapping conditioning variables $X_{i,t}$ into optimal portfolio weights $w_{i,t}$.

MSRR Neural Network Estimation

Model Structure

$$w_t = \beta' S_{t-1}(\theta)$$

$$1 = w_t' R_t + u_t$$

Training Objective (Nonlinear Least Squares)

$$\min_{\theta,\beta} \frac{1}{T} \sum_{t=1}^{T} \left(1 - \beta' S_{t-1}(\theta)' R_t\right)^2$$

Interpretation

- θ: learns nonlinear feature representation
- β: allocates capital across learned signals
- Joint optimization maximizes Sharpe
- **Distinction:** Portfolio-based loss $\neq$ return prediction loss $\min_\theta \sum (R_t - f(X_{t-1};\theta))^2$

From MSRR to Parametric Portfolio Policies (PPP)

Key Idea:
Instead of estimating expected returns, directly parameterize the portfolio rule.

$$w_{i,t} = f(x_{i,t}; \theta)$$

- $x_{i,t}$: observable firm characteristics
- θ: common parameters across assets

Crucial Difference from MSRR:

- MSRR maximizes Sharpe ratio (quadratic objective)
- PPP maximizes expected utility directly

Expected Utility Framework

Brandt et al. [2009] proposed:

$$\max_{\{w_{i,t}\}} E_t \left[u \left(\sum_{i=1}^{N_t} w_{i,t} r_{i,t+1} \right) \right]$$

Portfolio Rule:

$$w_{i,t} = f(x_{i,t}; \theta)$$

Estimation:

$$\max_{\theta} \frac{1}{T} \sum_{t=1}^{T} u \left(\sum_{i=1}^{N_t} f(x_{i,t}; \theta) r_{i,t+1} \right)$$

Key Features:

- Direct utility maximization
- Dynamic allocation
- No need to estimate μ or Σ

Linear Parametric Portfolio Policy (PPP)

Portfolio weights:

$$w_{i,t} = \bar{w}_{i,t} + \frac{1}{N_t} x_{i,t}' \theta$$

- $\bar{w}_{i,t}$: benchmark weight (e.g., equal-weight)
- $x_{i,t}$: firm characteristics
- θ: common parameter vector

Dimension Reduction:

$$N_t \gg K$$

Only estimate K parameters instead of N_t weights.

Deep Parametric Portfolio Policies (DPPP)

Simon et al. [2025] replace linear policy with a feed-forward neural network:

$$w_{i,t} = f(x_{i,t}; \theta)$$

where f is a deep neural network.

Training Objective:

$$\max_{\theta} \frac{1}{T} \sum_{t=1}^{T} u \left(\sum_{i=1}^{N_t} f(x_{i,t}; \theta) r_{i,t+1} \right)$$

Interpretation:

- End-to-end policy learning
- Nonlinear interaction between characteristics
- Direct optimization of economic objective

Empirical Evidence:

- DPPP improves Sharpe ratio from 1.8 (linear PPP) to 2.5.

Mean Variance Portfolio

Mean Variance Portfolio Implementation

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Considering trading costs

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